



Darshan
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Python Programming - 2301CS404

Lab - 5

Roll No.: 459

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1) Character Category Counter

Problem: Count uppercase, lowercase, digits, and special characters in a string.

Sample Inputs & Outputs:

- Input: Python3@Lab → Uppercase: 2, Lowercase: 7, Digits: 1, Special: 1
- Input: HELL0123 → Uppercase: 5, Lowercase: 0, Digits: 3, Special: 0
- Input: hi@#9 → Uppercase: 0, Lowercase: 2, Digits: 1, Special: 2

```
In [13]: str = input("Enter String : ")

lowercase , uppercase , digits , special = 0 , 0 , 0 , 0

for i in str:
    if(i.islower()):
        lowercase += 1
    elif(i.isupper()):
        uppercase += 1
    elif(i.isdigit()):
        digits += 1
    elif(not i.isalnum()):
        special += 1

print(f"Uppercase: {uppercase}, Lowercase: {lowercase}, Digits: {digits}, Special: {special}")
```

Enter String : hi@3#

Uppercase: 0, Lowercase: 2, Digits: 1, Special: 2

2) Reverse Each Word

- Input: Python is Easy → nohtyP si ysaE
- Input: Hello World → olleH dlrow
- Input: Learn Python Fast → nraeL nohtyP tsaF

```
In [20]: str = input("Enter String : ")

str = str[::-1]

li = str.split(" ")

li = li[::-1]

print(" ".join(li))
```

Enter String : Python is Easy
nohtyP si ysaE

3) Case Pattern Identifier

- PYTHON → Uppercase
- python → Lowercase
- Python Programming → Title Case

```
In [23]: str = input("Enter String : ")

if(str.islower()):
    print("LowerCase")
elif(str.isupper()):
    print("UpperCase")
elif(str.istitle()):
    print("TitleCase")
```

Enter String : py
LowerCase

4) First and Last Occurrence Finder

- programming , g → First:3 Last:10
- banana , a → First:1 Last:5
- hello , z → Not found

```
In [31]: str = input("Enter String : ")
ch = input("Enter ch : ")

if(str.find(ch) == -1):
    print("Not Found")
else:
    print(f"First : {str.find(ch)}") # index char no made to erro jyare find ch

    print(f"Last : {str.rfind(ch)}")

    # or here we also can use index and rindex
```

Enter String : programming
Enter ch : g

First : 3
Last : 10

5) Word Frequency Counter (ignore the case)

- Python is easy and Python is powerful → python: 2
- Java is popular but java is verbose → java: 2
- C is fast → c:1

```
In [35]: str = input("Enter String : ").lower()
word = input("Enter Word : ")

print(str.count(word))
```

Enter String : Python is easy and Python is powerful
Enter Word : python
2

6) Remove Extra Spaces

- Python is fun → Python is fun
- Hello World → Hello World
- NoExtraSpace → NoExtraSpace

```
In [52]: str = input("Enter String : ").strip()
li = str.split(" ")
li2 = []

for i in li:
    if(not (i == "")):
        li2.append(i)

print(" ".join(li2))
# print(li2)
```

Enter String : Hello World
Hello World

7) Prefix and Suffix Removal

- unhappy.txt → happy
- pretest.py → test
- unwanted.doc → wanted

```
In [ ]: s = input("Enter String : ")

prefixes = ["pre", "un"]
suffixes = [".txt", ".py", ".doc"]

# remove prefix
for p in prefixes:
    if s.startswith(p):
        s = s.removeprefix(p)
        break
```

```
# remove suffix
for suf in suffixes:
    if s.endswith(suf):
        s = s.removesuffix(suf)
        break

print(s)
```

Enter String : pretty.txt
tty

8) Replace Vowels

- Programming → Pr*gr*mm*ng
- Education → *d*c*t**n
- Sky → Sky

```
In [61]: str = input("Enter String : ")

for i in str:
    if(i.lower() in "aeiou"):
        str = str.replace(i,"*")

print(str)
```

Enter String : Programming
Pr*gr*mm*ng

9) String Compression

- aaabbccccc → a3b2c4
- xxxyyyz → x3y3z1
- abcd → a1b1c1d1

```
In [3]: st = input("Enter String : ")
s = ""
# for i in st:
#     if(s.find(i) == -1):
#         s += i + str(st.count(i))

for ch in st:

    if(ch not in s):
        s += ch + str(st.count(ch))

print(s)
```

a3b3c2

10) Toggle Case (without using str.swapcase())

- PyThOn → pYtHoN
- HELlo → heLLo
- Python → pYTHON

```
In [ ]: str = input("Enter String : ")

# print(str.swapcase())

s = ""

for i in str:
    if(i.isupper()):
        s += i.lower()
    elif(i.islower()):
        s += i.upper()
    else:
        s += i

print(s)
```

Enter String : PyThOn
pYtHoN

11) Username Validation

valid username rules:

1. only alphabets and digits
2. does not start with a digit
3. length ≥ 6

- User123 → Valid
- 123User → Invalid
- Us@12 → Invalid

```
In [ ]: username = input("Enter UserName : ")

if(username.isalnum() and not username[0].isdigit() and len(username) >= 6):
    print(f"{username} Is Valid")
else:
    print(f"{username} Is Invalid")
```

Enter UserName : Us@12
Us@12 Is Invalid

12) Palindrome Checker (ignore the case)

- Madam → Palindrome
- Level → Palindrome
- Python → Not Palindrome

```
In [ ]: word = input("Enter Word : ")

if(word.lower() == word[::-1].lower()):
    print(f"{word} Is Palindrome")
else:
    print(f"{word} Is Not Palindrome")
```

Enter Word : Python
Python Is Not Palindrome

13) Longest Word Finder

- Python programming is interesting → programming
- I love coding → coding
- Data structures and algorithms → structures

```
In [ ]: s = input("Enter Sentence : ")

li = s.split(" ")

ans = ""

for i in li:
    if(len(ans) < len(i)):
        ans = i

print("Longest Word Is",ans)
```

Enter Sentence : Data structures and algorithms
Longest Word Is structures

14) Case Conversion

- python programming
 - Upper: PYTHON PROGRAMMING
 - Lower: python programming
 - Title: Python Programming
 - Capitalize: Python programming

```
In [ ]: s = input("Enter Sentence : ")

print("Upper : ",s.upper())
print("Lower : ",s.lower())
print("Title : ",s.title())
print("Capitalize : ",s.capitalize())
```

Enter Sentence : python programming
Upper : PYTHON PROGRAMMING
Lower : python programming
Title : Python Programming
Capitalize : Python programming

15) Custom Split

- apple,banana,grape
- 10|20|30
- A-B-C-D

```
In [ ]: data = input("Enter Data : ")
sp = input("Split By : ")

print(data.split(sp))
```

```
Enter Data : A-B-C-D  
Split By : -  
['A', 'B', 'C', 'D']
```